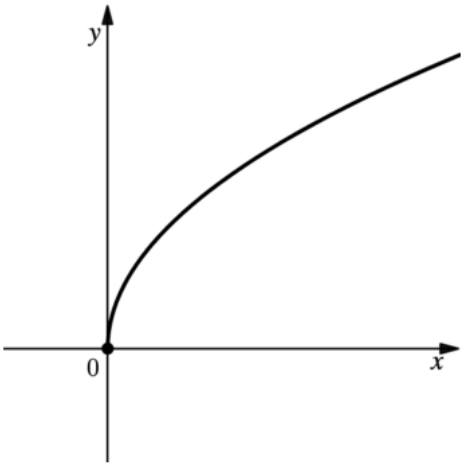
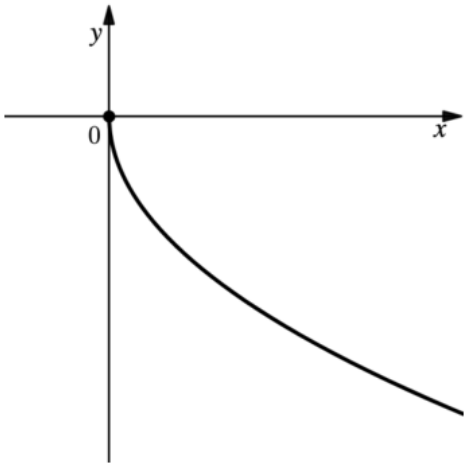
1.19 Graph Sketching: Radical Functions

Worked Solutions

1 a)



1 b)



2)

$7 = k\sqrt{4} \Rightarrow 7 = 2k \Rightarrow k = \frac{7}{2}$ so $y = \frac{7}{2}\sqrt{x}$

3 a)

$y = \sqrt{x-2}$

3 b)

$y = \sqrt{x+3}$

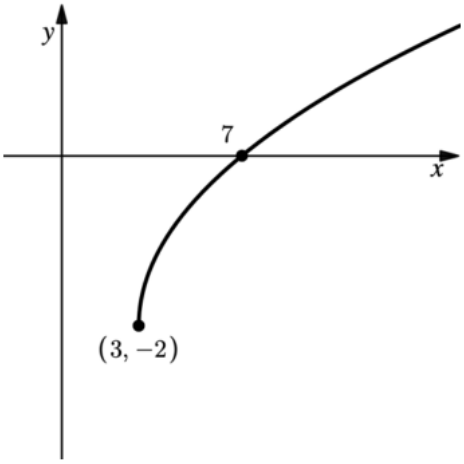
4 a)

$y = \sqrt{x+5} - 3$

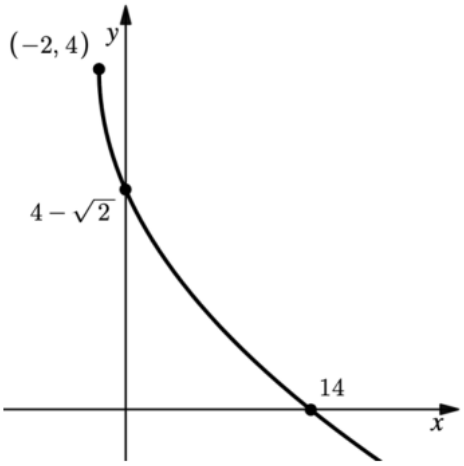
4 b)

$y = \sqrt{x+2} - \sqrt{2}$

5 a)



5 b)



6)

$4 - \sqrt{x} = \sqrt{x+4} \Rightarrow (4 - \sqrt{x})^2 = x + 4$

$\Rightarrow 16 - 8\sqrt{x} + x = x + 4 \Rightarrow 8\sqrt{x} = 12$

$\Rightarrow \sqrt{x} = \frac{3}{2} \Rightarrow x = \frac{9}{4}$

$y = 4 - \sqrt{\frac{9}{4}} = 4 - \frac{5}{2} = \frac{5}{2}$

So the graphs intersect at $(\frac{9}{4}, \frac{5}{2})$